Quartic Analysis

Sina Tootoonian, Claude

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We're given

- $\mathbf{X}$: an N cells $\times$ P odours matrix of observed input responses,
- $\mathbf{C}$: a P cells $\times$ P cells observed output covariance matrix,
- The regularization parameter $\lambda = \lambda_0 \frac{P^2}{N}$, where λ_0 is the value we sweep in the paper.
- The mean subtraction operation, $\mathbf{J} = \mathbf{I} - N^{-1}\mathbf{1}\mathbf{1}^T$.
- The prediction error $\mathbf{E}(\mathbf{z}) = \mathbf{X}^T[\mathbf{z}]\mathbf{J}[\mathbf{z}]\mathbf{X} - \mathbf{C}$

Our loss function is

$$L(\mathbf{z}) = \frac{1}{2}\|\mathbf{E}(\mathbf{z})\|_F^2 + \frac{\lambda}{2}\|\mathbf{z} - 1\|_2^2, \quad (1) \quad \texttt{\{eq:loss\}}$$

where [.] converts its argument to a diagonal matrix.

We want to view this from the perspective of a single unit z_i. To do that, we rewrite $\mathbf{E}(\mathbf{z})$ to isolate that unit. Examining its components,

$$\begin{aligned}
\mathbf{X}^T[\mathbf{z}]\mathbf{J}[\mathbf{z}]\mathbf{X} &= \mathbf{X}^T[\mathbf{z}^2]\mathbf{X} - N^{-1}\mathbf{X}^T\mathbf{z}\mathbf{z}^T\mathbf{X} \\
\mathbf{X}^T[\mathbf{z}^2]\mathbf{X} &= \sum_{j=1}^{N}\mathbf{x}_j\mathbf{x}_j^T z_j^2 = \mathbf{x}_i\mathbf{x}_i^T z_i^2 + \sum_{j\neq i}\mathbf{x}_j\mathbf{x}_j^T z_j^2 \\
\mathbf{m} &\triangleq \mathbf{X}^T\mathbf{z} = \sum_j \mathbf{x}_j\mathbf{z}_j = \mathbf{x}_i z_i + \sum_{j\neq i}\mathbf{x}_j\mathbf{z}_j = \mathbf{x}_i z_i + \mathbf{m}_i \\
\mathbf{E}(\mathbf{z}) &= \mathbf{x}_i\mathbf{x}_i^T z_i^2 + \sum_{j\neq i}\mathbf{x}_j\mathbf{x}_j^T z_j^2 - N^{-1}(\mathbf{x}_i z_i + \mathbf{m}_i)(\mathbf{x}_i z_i + \mathbf{m}_i)^T - \mathbf{C} \\
&= \mathbf{x}_i\mathbf{x}_i^T z_i^2 - N^{-1}(\mathbf{x}_i\mathbf{m}_i^T z_i + \mathbf{m}_i\mathbf{x}_i^T z_i + \mathbf{x}_i\mathbf{x}_i^T z_i^2) + \sum_{j\neq i}\mathbf{x}_j\mathbf{x}_j^T z_j^2 - N^{-1}\mathbf{m}_i\mathbf{m}_i^T - \mathbf{C} \\
&= \alpha\mathbf{x}_i\mathbf{x}_i^T z_i^2 - N^{-1}(\mathbf{x}_i\mathbf{m}_i^T + \mathbf{m}_i\mathbf{x}_i^T)z_i + \mathbf{E}_i.
\end{aligned}$$

Computing the goodness-of-fit term of the loss,

$$\begin{aligned}
\mathrm{tr}(\mathbf{E}(\mathbf{z})^T\mathbf{E}(\mathbf{z})) &= \alpha^2\|\mathbf{x}_i\|_2^4 z_i^4 - 4\alpha N^{-1}\|\mathbf{x}_i\|_2^2\mathbf{m}_i^T\mathbf{x}_i z_i^3 + 2\alpha\mathbf{x}_i^T\mathbf{E}_i\mathbf{x}_i z_i^2 \\
&\quad + 2N^{-2}z_i^2(\|\mathbf{x}_i\|_2^2\|\mathbf{m}_i\|_2^2 + (\mathbf{x}_i^T\mathbf{m}_i)^2) - 4N^{-1}z_i\mathbf{x}_i^T\mathbf{E}_i\mathbf{m}_i + \mathrm{tr}(\mathbf{E}_i^T\mathbf{E}_i).
\end{aligned}$$

Our loss function in terms of z_i is then

$$
\begin{aligned}
L(z_i) = &\frac{1}{2}\alpha^2\|\mathbf{x}_i\|_2^4 z_i^4 \\
&- \frac{2\alpha}{N}\|\mathbf{x}_i\|_2^2 \mathbf{m}_i^T \mathbf{x}_i z_i^3 \\
&+ \left(\alpha \mathbf{x}_i^T \mathbf{E}_i \mathbf{x}_i + \frac{\|\mathbf{x}_i\|_2^2\|\mathbf{m}_i\|_2^2 + (\mathbf{x}_i^T\mathbf{m}_i)^2}{N^2} + \frac{\lambda}{2}\right) z_i^2 \\
&- \frac{2\mathbf{x}_i^T\mathbf{E}_i\mathbf{m}_i + N\lambda}{N} z_i \\
&+ \frac{\mathrm{tr}(\mathbf{E}_i^T\mathbf{E}_i) + \lambda(1 + \sum_{i\neq j}(z_j - 1)^2)}{2}.
\end{aligned}
$$

The trailing constant doesn't matter for the minimization, so we'll drop it. We'll express the remaining coefficients in terms of a few geometric quantities:

- The raw power of the population mean $m \triangleq \mathbf{m}_i^T\mathbf{m}_i$;
- The raw channel power, $p_i \triangleq \mathbf{x}_i^T\mathbf{x}_i$;
- The residual channel power, $q_i \triangleq \mathbf{x}_i^T\mathbf{E}_i\mathbf{x}_i$;
- The raw alignment with the population mean, $a_i \triangleq \mathbf{x}_i^T\mathbf{m}_i$;
- The residual alignment with the population mean, $b_i \triangleq \mathbf{x}_i^T\mathbf{E}_i\mathbf{m}_i$;
- The curvature at the origin, $\kappa_i \triangleq \alpha q_i + \frac{p_i m + a_i^2}{N^2}$.

In these terms, and dropping the constant,

$$
\begin{aligned}
L(z_i) &= \frac{1}{2}\alpha^2 p_i^2 z_i^4 - \frac{2\alpha}{N}p_i a_i z_i^3 + \left(\kappa_i + \frac{\lambda}{2}\right) z_i^2 - \left(\frac{2b_i}{N} + \lambda\right) z_i && (2) \quad \{\texttt{eq:Lzi}\} \\
&\triangleq A z_i^4 + B z_i^3 + C z_i^2 + D z_i. && (3) \quad \{\texttt{eq:Lzi}\}
\end{aligned}
$$

To better understand this quartic, we will change coordinates. First, we'll rescale by A,

$$
\begin{aligned}
L(z) &\propto z^4 + \frac{B}{A}z^3 + \frac{C}{A}z^2 + \frac{D}{A}z \\
&\triangleq z^4 + bz^3 + cz^2 + dz.
\end{aligned}
$$

Next, we'll shift to a coordinate $s = z - \overline{z}$ in which the cubic disappears. Expressing the loss as a function $\tilde{L}$ of s, we note that for a monic quartic with no cubic, the third derivative should be $24s$.

$$
\begin{aligned}
\tilde{L}(s) &= (s+\overline{z})^4 + b(s+\overline{z})^3 + c(s+\overline{z})^2 + d(s+\overline{z}). \\
\tilde{L}'''(s) &= 24(s+\overline{z}) + 6b = 24s \implies \overline{z} = -\frac{b}{4}.
\end{aligned}
$$

So, in the shifted and scaled coordinates, the loss function is

$$
\tilde{L}(s) = \left(s - \frac{b}{4}\right)^4 + c\left(s - \frac{b}{4}\right)^2 + d\left(s - \frac{b}{4}\right)
$$

We can expand out the first equation to get the coefficients. But an easier way is to Taylor expand our loss function $L(z)$ to get a polynomial in s:

$$
\tilde{L}(s) = L(s+\overline{z}) = L(\overline{z}) + L'(\overline{z})s + \frac{1}{2}L''(\overline{z})s^2 + \frac{1}{24}L'''(\overline{z})s^4.
$$

Then, $L''''(\overline{z}) = 24$. The other derivatives are

$$L'(\overline{z}) = 4\overline{z}^3 + 3b\overline{z}^2 + 2c\overline{z} + d = -\frac{b^3}{16} + 3\frac{b^3}{16} - \frac{bc}{2} + d = \frac{b^3}{8} - \frac{bc}{2} + d$$
$$L''(\overline{z}) = 12\overline{z}^2 + 6b\overline{z} + 2c = 12\frac{b^2}{16} - 6\frac{b^2}{4} + 2c = -\frac{3}{4}b^2 + 2c.$$

So we arrive at

$$\begin{aligned}\tilde{L}(s) &= s^4 + \left(c - \frac{3}{8}b^2\right)s^2 + \left(d - \frac{bc}{2} + \frac{b^3}{8}\right)s + \text{const} \\ &\triangleq s^4 + \tilde{c}s^2 + \tilde{d}s + \text{const}. \end{aligned} \tag{4}$$

We can see that $\tilde{L}$ combines a convex quartic with a parabola. If the parabola is also convex, so $\tilde{c} \geq 0$, then $\tilde{L}$ will also be convex, and have only only one root. If the parabola is concave (pointing down), so $\tilde{c} < 0$, then $\tilde{L}$ may have one or two roots, depending on the tilt imposed by the linear term.

To determine the location of the root(s), we set the derivative to 0:

$$0 = \tilde{L}'(s) = 4s^3 + 2\tilde{c}s + \tilde{d} = s^3 + \frac{\tilde{c}}{2}s + \frac{\tilde{d}}{4} \triangleq s^3 + gs + h.$$

To determine the locations of the roots, we use the triple angle formula:

$$4\cos^3(\theta) - 3\cos(\theta) - \cos(3\theta) = 0.$$

Let $s = K\cos(\theta)$. Then our derivative equation is

$$K^3\cos(\theta)^3 + Kg\cos(\theta) + h = 0.$$

Dividing out by $K^3/4$, we get

$$4\cos(\theta)^3 + \frac{4g}{K^2}\cos(\theta) + \frac{4h}{K^3} = 0.$$

Now if $g \leq 0$, we can set K such that

$$\frac{4g}{K^2} = -3 \implies K = 2\sqrt{|g|/3}.$$

Then we get

$$\cos(3\theta) = -\frac{4h}{K^3} = -\frac{4h}{K^2}\frac{1}{K} = -\frac{3h}{2|g|}\sqrt{\frac{3}{|g|}} = \frac{3h}{2g}\sqrt{\frac{3}{-g}} \triangleq u.$$

This equation will have solution if $|u| \leq 1$, equivalently

$$1 \geq u^2 = \frac{27h^2}{-4g^3} \iff 27h^2 \leq 4(-g)^3,$$

with equality determining the *cusp*. The solutions will satisfy

$$3\theta = \arccos(u) + 2\pi k \implies \theta = \frac{1}{3}\arccos(u) + \frac{2\pi k}{3}, \quad k \in \{0, 1, 2\},$$

corresponding to the two wells and one ridge.

Which k corresponds to the ridge, and which to the valleys? When $u = 0$, there is no tilt in the quartic and the wells will be symmetric around 0. In this case,

$$\theta = \frac{1}{3}\arccos(0) + \frac{2\pi k}{3} = \frac{\pi}{6} + \frac{2\pi k}{3} = \left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}\right\}.$$

Then

$$\cos(\theta) = \{\sqrt{3}/2, -\sqrt{3}/2, 0\}.$$

So, assuming this ordering holds at non-zero u, $k = 0$ gives the solution when the tilt is negative, $k = 1$ gives the solution for positive tilt, and $k = 2$ gives the ridge.

Now we're interested in how large the cosine term can get. As the tilt becomes more negative, the minimum at $\sqrt{3}/2$ will move to the right. How far forward will it go? Negative tilt corresponds to negative h and hence positive u since g is negative in this regime. Hence the maximally negative tilt, occuring whent he ridge is about disappear, will correspond to the $u = 1$ case. At that point

$$\theta = \frac{1}{3}\arccos(1) + \frac{2\pi k}{3} = \left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}$$

So then

$$\cos(\theta) = \{1, -1/2, -1/2\},$$

corresponding to a global minimum at 1 and an inflection point at $-1/2$.

So, when $|u| \leq 1$, we can represent the solution as

$$s^* = \frac{\sqrt{3}}{2}K\text{sign}(u)f(|u|), \quad f \in \left[1, \frac{2}{\sqrt{3}}\right].$$

Plugging in our expression for K simplifies this to

$$s^* = \sqrt{|g|}\ \text{sign}(u)f(|u|).$$

Now let's consider the case when $g > 0$. In that case, the relevant triple-angle formula is for sinh,

$$4\sinh^3(\theta) + 3\sinh(\theta) - \sinh(3\theta) = 0.$$

If we set $s = K\sinh(\theta)$, our depressed cubic becomes

$$K^3\sinh(\theta)^3 + Kg\sinh(\theta) + h = 0.$$

Dividing by $K^3/4$ this gives

$$4\sinh(\theta)^3 + \frac{4g}{K^2}\sinh(\theta) + \frac{4h}{K^3} = 0.$$

Setting

$$\frac{4g}{K^2} = 3 \implies K = 2\sqrt{g/3}$$

gives

$$\sinh(3\theta) = -\frac{4h}{K^3} = -\frac{4h}{K^2}\frac{1}{K} = -3\frac{4h}{4g}\frac{1}{K} = -\frac{3h}{2g}\sqrt{\frac{3}{g}} \triangleq v.$$

We then get $\theta = \frac{1}{3}\text{arcsinh}(v)$, so

$$s^* = K\sinh(\theta) = 2\sqrt{\frac{g}{3}}\sinh\left(\frac{1}{3}\text{arcsinh}(v)\right).$$

When v is small,

$$s^* \approx \frac{1}{3}Kv = -\frac{h}{g}.$$

When v is large,

$$s^* \approx -h^{1/3}.$$

Complexifying

Rather than treating each of these cases separately, we can view them all in a unified manner by consider the roots of our depressed cubic in the complex plane.

In that setting, the triple angle formula is

$$w^3 + w^{-3} = (w + w^{-1})^3 - 3(w + w^{-1}).$$

We now parameterize

$$s = \frac{K}{2}(w + w^{-1}).$$

Our cubic equation then becomes

$$\frac{K^3}{8}(w + w^{-1})^3 + \frac{Kg}{2}(w + w^{-1}) + h = 0.$$

Normalizing,

$$(w + w^{-1})^3 + \frac{4g}{K^2}(w + w^{-1}) + \frac{8h}{K^3} = 0.$$

Setting K so that

$$\frac{4g}{K^2} = -3 \implies K = \sqrt{\frac{-4g}{3}} = 2\sqrt{\frac{-g}{3}},$$

Making that substitution gives

$$w^3 + w^{-3} = -\frac{8h}{K^3} \implies w^3 + w^{-3} + \frac{8h}{K^3} = 0.$$

Then letting $W = w^3$, we have

$$W + \frac{1}{W} + \frac{8h}{K^3} = 0.$$

Multipling by W, we get

$$W^2 + 2\frac{4h}{K^3}W + 1 = 0.$$

Since the constant coefficient is 1, we have that that two roots of this equation are inverses of each other.

Defining

$$u \triangleq -\frac{4h}{K^3} = -\frac{4h}{K^2}\frac{1}{K} = \frac{3h}{2g}\sqrt{\frac{3}{-g}}, \quad u^2 = -\frac{27h^2}{4g^3},$$

we get

$$W = \frac{2u \pm \sqrt{4u^2 - 4}}{2} = u \pm \sqrt{u^2 - 1}.$$

Now each value of W will gives us three values of w. There are two values of W, so this will yield six total values of w. However, these can be put in three pairs, where each element of the pair is the inverse of the other. And s is invariant to inverses, so $s(w) = s(w^{-1})$, and we get three values of s as the roots of the cubic, as expected.

Let's consider the various cases.

Case 1: $g < 0$

First, let's assume $g < 0$, so K is real. In that case, $u^2 \geq 0$.

If $|u| \leq 1$, then the square-root is pure imaginary, so $|W| = 1$ and W is on the unit circle. So

$$W = e^{j(3\theta + 2\pi k)} \implies w = e^{j\theta_k}, \quad \theta_k \triangleq \theta + \frac{2\pi k}{3}, \quad k \in \{0, 1, 2\}.$$

This in turn means three real roots

$$\boxed{s_k = K\cos\left(\theta + \frac{2\pi k}{3}\right), \quad k \in \{0,1,2\}, \quad \theta = \frac{1}{3}\arccos(u).}$$

If $|u| > 1$, then the square-root in the definition of W is real, so

$$w = W^{1/3} \implies w_k = \sqrt[3]{W}e^{j2\pi k/3}.$$

Then

$$\begin{aligned} s_0 &= K\,\mathrm{sgn}(W)\frac{|W|^{1/3} + |W|^{-1/3}}{2} = K\,\mathrm{sgn}(W)\frac{e^{\frac{\ln(|W|)}{3}} + e^{-\frac{\ln(|W|)}{3}}}{2} \\ &= K\,\mathrm{sgn}(W)\cosh\left(\frac{\ln(|W|)}{3}\right) = K\,\mathrm{sgn}(W)\cosh\left(\frac{\operatorname{arccosh}|u|}{3}\right). \end{aligned}$$

The next root is

$$s_1 = K\,\mathrm{sgn}(W)\frac{|W|^{1/3}E[j2\pi/3] + |W|^{-1/3}E[-j2\pi/3]}{2}, \quad E[x] \triangleq e^{jx},$$

while

$$s_2 = K\,\mathrm{sgn}(W)\frac{|W|^{1/3}E[-j2\pi/3] + |W|^{-1/3}E[j2\pi/3]}{2} = \overline{s_1}.$$

Since the roots of a depressed cubic sum to zero, we get

$$s_{1,2} = -\frac{s_0}{2} \pm j\eta.$$

The imaginary component η should be easy to compute. Since

$$\mathrm{Im}(E[j2\pi/3]) = \frac{\sqrt{3}}{2} = -\mathrm{Im}(E[-j2\pi/3]),$$

$$\eta = \mathrm{Im}(s_1) = K\,\mathrm{sgn}(W)\frac{\sqrt{3}}{2}\frac{|W|^{1/3} - |W|^{-1/3}}{2} = \frac{\sqrt{3}}{2}K\,\mathrm{sgn}(W)\sinh\left(\frac{\operatorname{arccosh}|u|}{3}\right).$$

We therefore have, using that $\mathrm{sgn}(W) = \mathrm{sgn}(u)$,

$$\boxed{s_0 = K\,\mathrm{sgn}(u)\cosh(\phi), \quad s_{1,2} = \frac{K}{2}\mathrm{sgn}(u)(-\cosh(\phi) \pm j\sqrt{3}\sinh(\phi)), \quad \phi = \frac{1}{3}\operatorname{arccosh}(|u|).}$$

Case 2: $g > 0$

Next, let's assume $g > 0$, so K is imaginary. Then $u^2 < 0$, so u is imaginary, and W is pure imaginary, say ja. Since the product of it and its paired root jb is 1,

$$1 = jajb = -1ab \implies b = -1/a,$$

so the roots are $W \in \{ja, -j/a\}$. Each such pair will give the same s. So we can work with one element, and absorb signs into it so that $a > 0$. To ensure this condition holds, we have to take the right root of the quadratic definining W. That root ends up being the positive one: letting $u = j\nu$ for some real ν, we get that

$$u = j\nu, \quad W = ja, \quad a > 0 \implies a = \nu + \sqrt{\nu^2 + 1}.$$

Then since $W = ja = e^{\ln(a) + j(\pi/2 + 2\pi k)}$, we get

$$w = W^{1/3} = e^{\frac{1}{3}\ln(a) + j\theta_k}, \quad \theta_k = \frac{\pi}{6} + \frac{2\pi k}{3}, \quad k \in \{0,1,2\}.$$

When $k=0$, $\theta_0 = \frac{\pi}{6}$, so,

$$j(w_0 + w_0^{-1}) = \sqrt[3]{a}E[\pi/6 + \pi/2] + \sqrt[3]{a^{-1}}E[-\pi/6 + \pi/2] = \sqrt[3]{a}E[2\pi/3] + \sqrt[3]{a^{-1}}E[\pi/3].$$

When $k=1$, $\theta_1 = \frac{\pi}{6} + \frac{4\pi}{6} = \frac{5\pi}{6}$ so

$$j(w_1 + w_1^{-1}) = \sqrt[3]{a}E[5\pi/6 + \pi/2] + \sqrt[3]{a^{-1}}E[-5\pi/6 + \pi/2] = \sqrt[3]{a}E[-2\pi/3] + \sqrt[3]{a^{-1}}E[-\pi/3].$$

We therefore see that the corresponding roots are complex conjugates of each-other:

$$s_0 = \mathrm{Im}(K)\frac{\sqrt[3]{a}E[2\pi/3] + \sqrt[3]{a^{-1}}E[\pi/3]}{2}, \quad s_1 = \overline{s_0}.$$

For the remaining case of $k=2$, $\theta_2 = \frac{9\pi}{6} = 3\pi/2 = -\pi/2$, so

$$j(w_2 + w_2^{-1}) = \sqrt[3]{a}[-\pi/2 + \pi/2] + \sqrt[3]{a^{-1}}[\pi/2 + \pi/2] = \sqrt[3]{a} - \sqrt[3]{a^{-1}},$$

so

$$\frac{j(w_2 + w_2^{-1})}{2} = \frac{e^{\ln(a)/3} - e^{-\ln(a)/3}}{2} = \sinh(\ln(a)/3).$$

Since $\ln(\nu + \sqrt{\nu^2+1}) = \operatorname{arcsinh}(\nu)$, our expression above simplifies. Combining it with K, we get

$$s_2 = \mathrm{Im}(K)\sinh\left(\frac{1}{3}\operatorname{arcsinh}(\nu)\right).$$

Since the roots of a depressed cubic sum to zero, we get

$$s_{0,1} = -\frac{s_2}{2} \pm j\eta.$$

The imaginary component is easy to compute, since

$$\mathrm{Im}(E[2\pi/3]) = \mathrm{Im}(E[\pi/3]) = \frac{\sqrt{3}}{2}.$$

Then

$$\eta = \frac{\mathrm{Im}(K)}{2}\sqrt{3}\cosh\left(\frac{1}{3}\operatorname{arcsinh}(\nu)\right),$$

so that

$$\boxed{s_2 = \mathrm{Im}(K)\sinh(\phi), \quad s_{0,1} = \frac{\mathrm{Im}(K)}{2}(-\sinh(\phi) \pm j\sqrt{3}\cosh(\phi)), \quad \phi \triangleq \frac{1}{3}\operatorname{arcsinh}(\nu).}$$